

1. Administrative & Study Identification

Full Study Title: A Randomized Double-blind, Placebo-controlled, Multicenter Trial Assessing the Impact of Lipoprotein (a) Lowering With Pelacarsen (TQJ230) on Major Cardiovascular Events in Patients With Established Cardiovascular Disease **Short Title/ Acronym:** Lp(a)HORIZON Trial **Protocol Number:** CTQJ230A12301 / 2019-001076-11 **NCT Number:** NCT04023552 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** Pelacarsen (TQJ230) **Phase of Development:** Phase III **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To demonstrate the superiority of pelacarsen (TQJ230) compared to placebo in reducing the risk of the expanded Major Adverse Cardiovascular Events (MACE) composite of cardiovascular (CV) death, non-fatal myocardial infarction (MI), non-fatal stroke, and urgent coronary re-vascularization requiring hospitalization. **Secondary Objectives:** To evaluate the effect of pelacarsen on the incidence of coronary heart disease (CHD) outcomes (death due to CHD, non-fatal MI, and urgent coronary revascularization) and percent reduction in Lp(a) levels.

2.2 Background & Rationale Elevated Lipoprotein(a) [Lp(a)] is an independent, genetic, and causal risk factor for cardiovascular disease (CVD) and calcific aortic valve stenosis. Currently, there are no approved pharmacological therapies to effectively lower Lp(a). Because Lp(a) levels cannot be adequately controlled through improved diet, increased physical activity, or existing lipid-lowering therapies (like statins), this study aims to demonstrate that lowering Lp(a) with the antisense oligonucleotide pelacarsen reduces cardiovascular morbidity and mortality in patients with established CVD.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** 1:1 (Pelacarsen vs. Placebo)

3.2 Planned Enrollment & Duration Target **SN\$:** 8,323 participants **Number of Sites:** Global, multicenter (Argentina + 45 other countries) **Estimated Study Start:** 11-Dec-2019 **Estimated Primary**

Completion: May-2025 (overall duration anticipated to be ~6 years)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. History of established CVD defined as a myocardial infarction (MI) or ischemic stroke (between 3 months and 10 years ago), or symptomatic peripheral artery disease. 2. Elevated Lp(a) levels of ≥ 70 mg/dL (approximately 149 nmol/L) at screening, as confirmed by a central lab. 3. Receiving optimized standard of care therapy for CVD (e.g., statins, PCSK9 inhibitors).

4.2 Exclusion Criteria 1. Active liver disease or severe kidney issues. 2. Uncontrolled blood pressure, advanced heart failure, or history of low platelet count. 3. Recent history of major bleeding or active cancer.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Pelacarsen 80 mg administered monthly (every 4 weeks). **Route of Administration:** Subcutaneous (SQ) injection. **Duration of Treatment:** Event-driven study with a required minimum follow-up of 2.5 years.

5.2 Concomitant & Prohibited Therapy Permitted: Background optimized standard-of-care therapy for CVD, including LDL-C-lowering therapies such as statins or PCSK9 inhibitors. **Prohibited:** Other investigational drugs or non-approved therapies targeting Lp(a) lowering.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -14 to 0 Informed Consent, Medical History, Lp(a) Central Lab Testing
Baseline	Day 0	Randomization (1:1), First Subcutaneous Dose	Interim Every 4 Weeks Investigational Product Administration (SQ), Efficacy Assessment, Safety Labs
End of Study	Event-Driven	Final Vital Signs, Exit Interview, IP Accountability, MACE Event Adjudication	

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Time to the first occurrence of the composite endpoint of major adverse cardiovascular events (CV death, non-fatal MI, non-fatal stroke, and urgent coronary re-vascularization requiring hospitalization). **Measurement Method:**

Clinical Endpoint Committee (CEC) confirmed events. Evaluated in both the overall population (≥ 70 mg/dL) and a subpopulation of Lp(a) ≥ 90 mg/dL.

7.2 Secondary & Exploratory Endpoints - Time to the first occurrence of the CEC confirmed composite endpoint of coronary heart disease (CHD death, non-fatal MI, urgent coronary re-vascularization requiring hospitalization). - Percent change in Lp(a) levels from baseline.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine collection via site interviews, laboratory tracking (including liver function and platelet counts), and physical examinations. **Serious Adverse Events (SAEs):** Standard reporting timeline (typically 24 hours). **Data Monitoring Committee (DMC):** Independent Clinical Endpoint Committee (CEC) for MACE adjudication and an external Data Safety Monitoring Board.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy; Safety Set for AE/SAE evaluations. Primary efficacy incorporates multiplicity control to test endpoints in both the overall population and the ≥ 90 mg/dL subpopulation. **Sample Size Justification:** Target enrollment of ~8,323 patients to accrue 993 CEC-confirmed primary CV events, ensuring adequate statistical power for the superiority hypothesis. **Handling of Missing Data:** Standard survival analysis methods (e.g., Kaplan-Meier, Cox proportional hazards) censoring for dropouts and missing time-to-event data.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring for drug accountability, AE review, and trial compliance. **Audit Trail:** Assured via eCRF locking, query resolution processes, and clinical event adjudication logs.

11. Study Identifiers & Governance

Primary ID: CTQJ230A12301 **Registry Number:** NCT04023552 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** Lp(a)HORIZON Trial **Official Regulatory Title:** A Randomized Double-blind, Placebo-controlled, Multicenter Trial Assessing the

12. Clinical Framework

Primary Condition: Cardiovascular Disease (CVD) **Diagnosis Threshold:** \$Lp(a) \ge 70\$ mg/dL (approx. 149 nmol/L) **Medical Methodology:** Antisense oligonucleotide (ASO) directed against apolipoprotein(a) production in the liver **Optimization Target:** Reduction in circulating Lipoprotein(a) and associated MACE risk

13. Study Procedures & Methodology

Management Pathway: Monthly SQ pelacarsen injections administered on top of maximally tolerated standard-of-care CVD treatment. **Education Protocol (HCP):** Training on SQ injection protocols, AE monitoring (specifically injection site reactions and lab anomalies), and event reporting. **Education Protocol (Patient):** Education on treatment adherence, recognizing signs of MACE (MI, stroke), and routine follow-ups. **Quality Audit Mechanism:** Routine source data verification (SDV) for CEC event packages and central laboratory sample handling.

14. Observation & Follow-Up Schedule

Total Duration: Event-driven study (minimum 2.5 years, estimated ~6 years total duration) **Onsite Visit Cadence:** Monthly (Every 4 weeks for drug administration and evaluation) **Remote Monitoring Frequency:** As per study clinical monitoring plan. **Checklist Review Interval:** Routine intervals aligning with the monthly clinic visits.

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				